

The ambient representation of the fixed algebra

`MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleRepresentation`

def

Defines the inclusion representation $\rho_0 : T \rightarrow B(\mathcal{H})$.

Statement

Let A be the completed CAR algebra and $T = C^*(\pi(A) \cup \{L_i : i \in I\}) \subseteq B(\mathcal{H})$ the algebra generated by the selected direct sum π of pure-state GNS representations indexed by their unitary-equivalence classes $i \in I$ and the unitaries L_i from the fixed homogeneity and shell construction. Write $\iota(a) = \pi(a)$ for the embedding $A \rightarrow T$. The inclusion representation is

$$\rho_0 : T \longrightarrow B(\mathcal{H}), \quad \rho_0(t)x = t(x).$$

Definition

The map ρ_0 is the literal inclusion: $\rho_0(t) = t$ as a bounded operator on $\mathcal{H}$. The source map $\iota(a)$ is $\pi(a)$ regarded as an element of T , so $\rho_0 \circ \iota = \pi$.

Assumptions

The GNS direct sum, the unitaries L_i and the generated algebra T are fixed as above. Each $t \in T$ is already a bounded operator on $\mathcal{H}$.

Conclusion

ρ_0 is a unital $*$ -representation and $\rho_0 \circ \iota = \pi$. Its definition introduces no new choice of Hilbert space or operators.

Faithfulness and irreducibility of this particular inclusion are separate theorems cited below. They are not additional data chosen in the definition; no separability of $\mathcal{H}$ is assumed.

Main citations

- Definition and its exact construction —
`MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleRepresentation`
- The inclusion before specialization — `MathlibAnnex.CStarAlgebra.CAR.shellFamilyInclusion`
- The CAR embedding into the same generated algebra —
`MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleSourceHom`
- Faithfulness of the inclusion — `MathlibAnnex.CStarAlgebra.CAR.shellFamilyInclusion_injective`
- Irreducibility of the inclusion —
`MathlibAnnex.CStarAlgebra.CAR.isIrreducible_shellFamilyInclusion`

Supporting route explanation

Lean source signature (exact)

```
noncomputable def atomicCounterexampleRepresentation :  
  Representation AtomicCounterexampleAlgebra  
    (MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState) :=  
  shellFamilyInclusion homogeneityShellFamily
```

In the source	Mathematical meaning
AtomicCounterexampleAlgebra	The one fixed algebra $T = C^*(\pi(A), \{L_i : i \in I\}) \subseteq B(\mathcal{H})$ supplied by the CAR homogeneity shell construction.
SelectedAtomicHilbert completedRootPureState	The selected atomic Hilbert sum $\mathcal{H} = \bigoplus_{i \in I} H_i$ based at the bundled root pure state ω . It is not assumed separable.
Representation AtomicCounterexampleAlgebra (...)	The output is the unital star representation $\rho_0 : T \rightarrow B(\mathcal{H})$.
shellFamilyInclusion homogeneityShellFamily	The complete RHS is the literal inclusion for the same fixed family: $\rho_0(t)x = t(x)$. It is distinct from the later trace-model action on H_τ . Faithfulness and irreducibility are separate theorems, not data chosen here.

Further source notes: SelectedAtomicHilbert completedRootPureState is $\mathcal{H}$; completedRootPureState is ω bundled with its purity proof. The map ι is explained through the separately cited source-map declaration.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleRepresentation
Accepted content SHA-256: e3afaf04cc08e85ac394ecbf13dd409b7a55543224dc769f0e8a5d20c4548b0b
Accepted source guide SHA-256: dcee1d5b2df53f811ca47b8374102e814f95ea0fc3002417b733b2766363585b
Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73